

# **Mathematics Bootcamp**

## **Introduction to Proofs and Choice Theory**

Miguel Vázquez-Carrero

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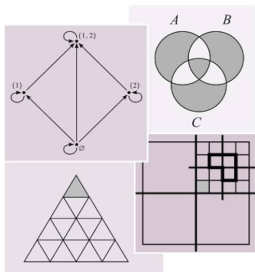
# Logic and Precision

- From Ordinary Language to Scientific Language.
- Symbols, Ideas and the Real world.
- Ideas, Truth Value and Obscurantism.

## These Slides

- In general, I will use the slides to save time and keep notation for the theorems: straight book screenshots.
- But most of the work will be in the board.  
**We learn a language by practice:** trial and error.
- Some believe that studying measure theory is a great way to start. This is like syntactic analysis for a baby in his first words.

# Just Prove It.



- It is important to reflect on the fact that these statements have **truth value**, i.e. they could be true or false.
- With the elements of logic, we can justify its truth-value.
- Most of what follows is taken from Velleman's book.

# Structured Programming

do

if [condition]

[List of instructions goes here.]

else

[Alternative list of instructions goes here.]

while [condition]

# Proofs as Structured Programming

Let  $x$  be arbitrary.

Suppose  $P(x)$  is true.

[Proof of  $Q(x)$  goes here.]

Thus, if  $P(x)$  then  $Q(x)$ .

Thus, for all  $x$ , if  $P(x)$  then  $Q(x)$ .

# Logical Connectives

**Conjunction:**  $P \wedge Q$

| $P$ | $Q$ | $P \wedge Q$ |
|-----|-----|--------------|
| $T$ | $T$ | $T$          |
| $T$ | $F$ | $F$          |
| $F$ | $T$ | $F$          |
| $F$ | $F$ | $F$          |

**Disjunction:**  $P \vee Q$

| $P$ | $Q$ | $P \vee Q$ |
|-----|-----|------------|
| $T$ | $T$ | $T$        |
| $T$ | $F$ | $T$        |
| $F$ | $T$ | $T$        |
| $F$ | $F$ | $F$        |

**Negation:**  $\neg P$

| $P$ | $\neg P$ |
|-----|----------|
| $T$ | $F$      |
| $F$ | $T$      |

# Quantifiers and De Morgan's Laws

## De Morgan's laws

$\neg(P \wedge Q)$  is equivalent to  $\neg P \vee \neg Q$ .

$\neg(P \vee Q)$  is equivalent to  $\neg P \wedge \neg Q$ .

## Quantifier Negation laws

$\neg \exists x P(x)$  is equivalent to  $\forall x \neg P(x)$ .

$\neg \forall x P(x)$  is equivalent to  $\exists x \neg P(x)$ .



## 1. Implication: $P \rightarrow Q$

| $P$ | $Q$ | $P \rightarrow Q$ |
|-----|-----|-------------------|
| $T$ | $T$ | $T$               |
| $T$ | $F$ | $F$               |
| $F$ | $T$ | $T$               |
| $F$ | $F$ | $T$               |

## 2. Contrapositive: $\neg Q \rightarrow \neg P$

| $P$ | $Q$ | $\neg Q$ | $\neg P$ | $\neg Q \rightarrow \neg P$ | $P \rightarrow Q$ |
|-----|-----|----------|----------|-----------------------------|-------------------|
| $T$ | $T$ | $F$      | $F$      | $T$                         | $T$               |
| $T$ | $F$ | $T$      | $F$      | $F$                         | $F$               |
| $F$ | $T$ | $F$      | $T$      | $T$                         | $T$               |
| $F$ | $F$ | $T$      | $T$      | $T$                         | $T$               |

### 3. Contradiction: $\neg(P \wedge \neg Q)$ **vs** $P \rightarrow Q$

| $P$ | $Q$ | $\neg Q$ | $P \wedge \neg Q$ | $\neg(P \wedge \neg Q)$ | $P \rightarrow Q$ |
|-----|-----|----------|-------------------|-------------------------|-------------------|
| $T$ | $T$ | $F$      | $F$               | $T$                     | $T$               |
| $T$ | $F$ | $T$      | $T$               | $F$                     | $F$               |
| $F$ | $T$ | $F$      | $F$               | $T$                     | $T$               |
| $F$ | $F$ | $T$      | $F$               | $T$                     | $T$               |

## 4. Proofs by Induction

**1.2.2 Principle of Mathematical Induction** *Let  $S$  be a subset of  $\mathbb{N}$  that possesses the two properties:*

1. *The number  $1 \in S$ .*
2. *For every  $k \in \mathbb{N}$ , if  $k \in S$ , then  $k + 1 \in S$ .*

*Then we have  $S = \mathbb{N}$ .*

**1.2.3 Principle of Mathematical Induction (second version)** *Let  $n_0 \in \mathbb{N}$  and let  $P(n)$  be a statement for each natural number  $n \geq n_0$ . Suppose that:*

1. *The statement  $P(n_0)$  is true.*
2. *For all  $k \geq n_0$ , the truth of  $P(k)$  implies the truth of  $P(k + 1)$ .*

*Then  $P(n)$  is true for all  $n \geq n_0$ .*

# From Logic to Set Theory

## Definition 1.4.1

The *intersection* of two sets  $A$  and  $B$  is the set  $A \cap B$  defined as follows:

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}.$$

The *union* of  $A$  and  $B$  is the set  $A \cup B$  defined as follows:

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}.$$

The *difference* of  $A$  and  $B$  is the set  $A \setminus B$  defined as follows:

$$A \setminus B = \{x \mid x \in A \text{ and } x \notin B\}.$$

# Set Theory

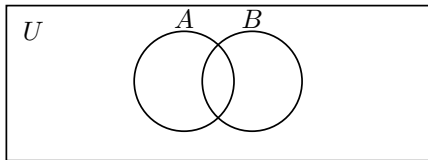


Figure 1.7.

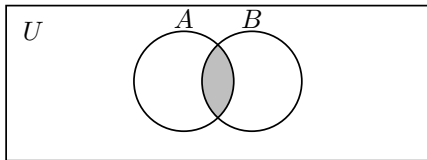


Figure 1.8.  $A \cap B$ .

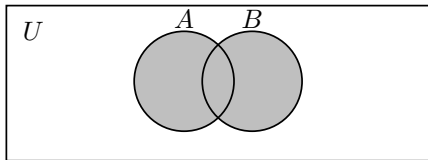


Figure 1.9.  $A \cup B$ .

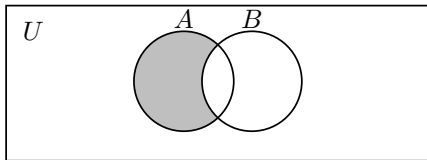


Figure 1.10.  $A \setminus B$ .

# Set Theory

## Definition 1.4.5

Suppose  $A$  and  $B$  are sets. We will say that  $A$  is a *subset* of  $B$  if every element of  $A$  is also an element of  $B$ . We write  $A \subseteq B$  to mean that  $A$  is a subset of  $B$ .  $A$  and  $B$  are said to be *disjoint* if they have no elements in common. Note that this is the same as saying that the set of elements they have in common is the empty set, or in other words  $A \cap B = \emptyset$ .

- Given a set  $A \subseteq U$ , define the *complement*  $A^C$
- De Morgan's Laws also apply in Sets:

$$(A \cup B)^c = A^c \cap B^c,$$

$$(A \cap B)^c = A^c \cup B^c.$$

# Set Theory

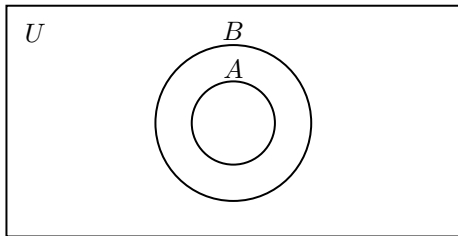


Figure 1.14.  $A \subseteq B$ .

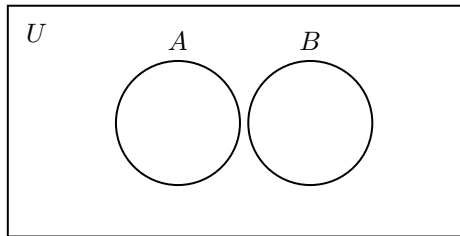


Figure 1.15.  $A \cap B = \emptyset$ .



# Set Theory

## Definition 4.1.1

Suppose  $A$  and  $B$  are sets. Then the *Cartesian product* of  $A$  and  $B$ , denoted  $A \times B$ , is the set of all ordered pairs in which the first coordinate is an element of  $A$  and the second is an element of  $B$ . In other words,

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}.$$

# Binary Relations

## Definition 4.2.1

Suppose  $A$  and  $B$  are sets. Then a set  $R \subseteq A \times B$  is called a *relation from  $A$  to  $B$* .

## Definition 4.3.2

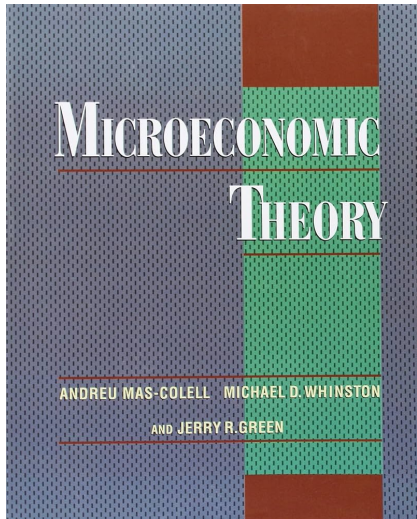
Suppose  $R$  is a relation on  $A$ .

1.  $R$  is said to be *reflexive on  $A$*  (or just *reflexive*, if  $A$  is clear from context) if  $\forall x \in A (xRx)$ , or in other words  $\forall x \in A ((x, x) \in R)$ .
2.  $R$  is *symmetric* if  $\forall x \in A \forall y \in A (xRy \rightarrow yRx)$ .
3.  $R$  is *transitive* if  $\forall x \in A \forall y \in A \forall z \in A ((xRy \wedge yRz) \rightarrow xRz)$ .

# Time for Economics

- We have seen some mathematical objects and their properties.
- More importantly, these are the words in the formal language that we use in Economics.
- Let's try to read some Economics.

# MWG: Microeconomic Theory



- Direct Screenshots:  
*Feel* that you can learn from it.
- This should not be your unique resource.
- In my 1st year, I also used:
  - **Levin Teaching Notes**
  - Rubinstein Notes Book
  - Kreps Books: 1990, 2012

## Preference and Choice

### 1.A Introduction

In this chapter, we begin our study of the theory of individual decision making by considering it in a completely abstract setting. The remaining chapters in Part I develop the analysis in the context of explicitly economic decisions.

The starting point for any individual decision problem is a *set of possible (mutually exclusive) alternatives* from which the individual must choose. In the discussion that follows, we denote this set of alternatives abstractly by  $X$ .

# Preference-Based Approach

## 1.B Preference Relations

In the preference-based approach, the objectives of the decision maker are summarized in a *preference relation*, which we denote by  $\succeq$ . Technically,  $\succeq$  is a binary relation on the set of alternatives  $X$ , allowing the comparison of pairs of alternatives  $x, y \in X$ . We read  $x \succeq y$  as “ $x$  is at least as good as  $y$ .” From  $\succeq$ , we can derive two other important relations on  $X$ :

- (i) The *strict preference* relation,  $\succ$ , defined by

$$x \succ y \Leftrightarrow x \succeq y \text{ but not } y \succeq x$$

and read “ $x$  is preferred to  $y$ .”<sup>1</sup>

- (ii) The *indifference* relation,  $\sim$ , defined by

$$x \sim y \Leftrightarrow x \succeq y \text{ and } y \succeq x$$

and read “ $x$  is indifferent to  $y$ .”

## Preference-Based Approach

In much of microeconomic theory, individual preferences are assumed to be *rational*. The hypothesis of rationality is embodied in two basic assumptions about the preference relation  $\succsim$ : *completeness* and *transitivity*.<sup>2</sup>

**Definition 1.B.1:** The preference relation  $\succsim$  is *rational* if it possesses the following two properties:

- (i) *Completeness*: for all  $x, y \in X$ , we have that  $x \succsim y$  or  $y \succsim x$  (or both).
- (ii) *Transitivity*: For all  $x, y, z \in X$ , if  $x \succsim y$  and  $y \succsim z$ , then  $x \succsim z$ .

- *Do not waste any attention* speculating about rationality (yet!).
- **Assume** a DM with rational preferences and see where it leads...

**This is exactly the approach we follow.**

# Preference-Based Approach

## 1.B Preference Relations

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# Preference-Based Approach

**Proposition 1.B.1:** If  $\succsim$  is rational then:

- (i)  $\succ$  is both *irreflexive* ( $x \succ x$  never holds) and *transitive* (if  $x \succ y$  and  $y \succ z$ , then  $x \succ z$ ).
- (ii)  $\sim$  is *reflexive* ( $x \sim x$  for all  $x$ ), *transitive* (if  $x \sim y$  and  $y \sim z$ , then  $x \sim z$ ), and *symmetric* (if  $x \sim y$ , then  $y \sim x$ ).
- (iii) if  $x \succ y \succsim z$ , then  $x \succ z$ .

- We can prove this proposition. Let's do it.

# Choice-Based Approach

## 1.C Choice Rules

In the second approach to the theory of decision making, choice behavior itself is taken to be the primitive object of the theory. Formally, choice behavior is represented by means of a *choice structure*. A choice structure  $(\mathcal{B}, C(\cdot))$  consists of two ingredients:

(i)  $\mathcal{B}$  is a family (a set) of nonempty subsets of  $X$ ; that is, every element of  $\mathcal{B}$  is a set  $B \subset X$ . By analogy with the consumer theory to be developed in Chapters 2 and 3, we call the elements  $B \in \mathcal{B}$  *budget sets*.

(ii)  $C(\cdot)$  is a *choice rule* (technically, it is a correspondence) that assigns a nonempty set of chosen elements  $C(B) \subset B$  for every budget set  $B \in \mathcal{B}$ . When  $C(B)$  contains a single element, that element is the individual's choice from among the alternatives in  $B$ . The set  $C(B)$  may, however, contain more than one element. When it does, the elements of  $C(B)$  are the alternatives in  $B$  that the decision maker *might* choose; that is, they are her *acceptable alternatives* in  $B$ . In this case, the set  $C(B)$

# Choice-Based Approach

**Definition 1.C.1:** The choice structure  $(\mathcal{B}, C(\cdot))$  satisfies the *weak axiom of revealed preference* if the following property holds:

If for some  $B \in \mathcal{B}$  with  $x, y \in B$  we have  $x \in C(B)$ , then for any  $B' \in \mathcal{B}$  with  $x, y \in B'$  and  $y \in C(B')$ , we must also have  $x \in C(B')$ .

**Definition 1.C.2:** Given a choice structure  $(\mathcal{B}, C(\cdot))$  the *revealed preference relation*  $\succsim^*$  is defined by

$x \succsim^* y \Leftrightarrow$  there is some  $B \in \mathcal{B}$  such that  $x, y \in B$  and  $x \in C(B)$ .